



EUROPEAN
SPALLATION
SOURCE

ESS

A superconducting proton linac

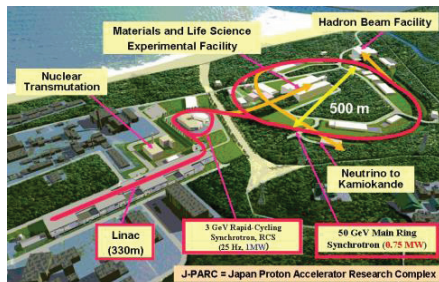
Mats Lindroos, David McGinnis, Tom Shea and Mohammad Eshraqi on
behalf of the ESS

Pasadena, NA PAC, 4 October 2013

Why ESS ?



- One of Europe's largest planned research infrastructure
- Will bring new insights to the grand challenges of science and innovation
- Collaborative project: Includes more than 17 countries
- 2014: Ground breaking
- 2019: Provide the world's most advanced tools for studying materials with neutrons **co-located with a very bright light source (MAX-IV)**

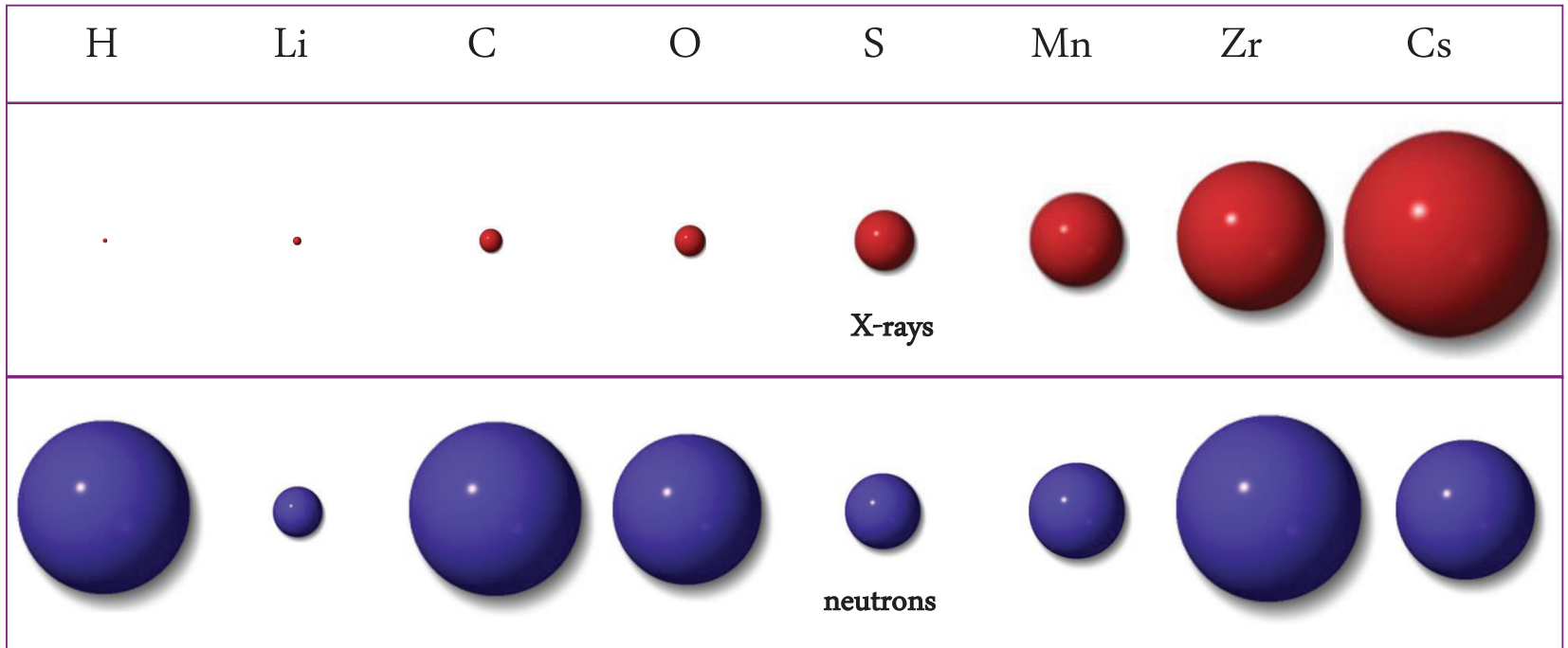


Japan 2008:
JPARC (<1MW)

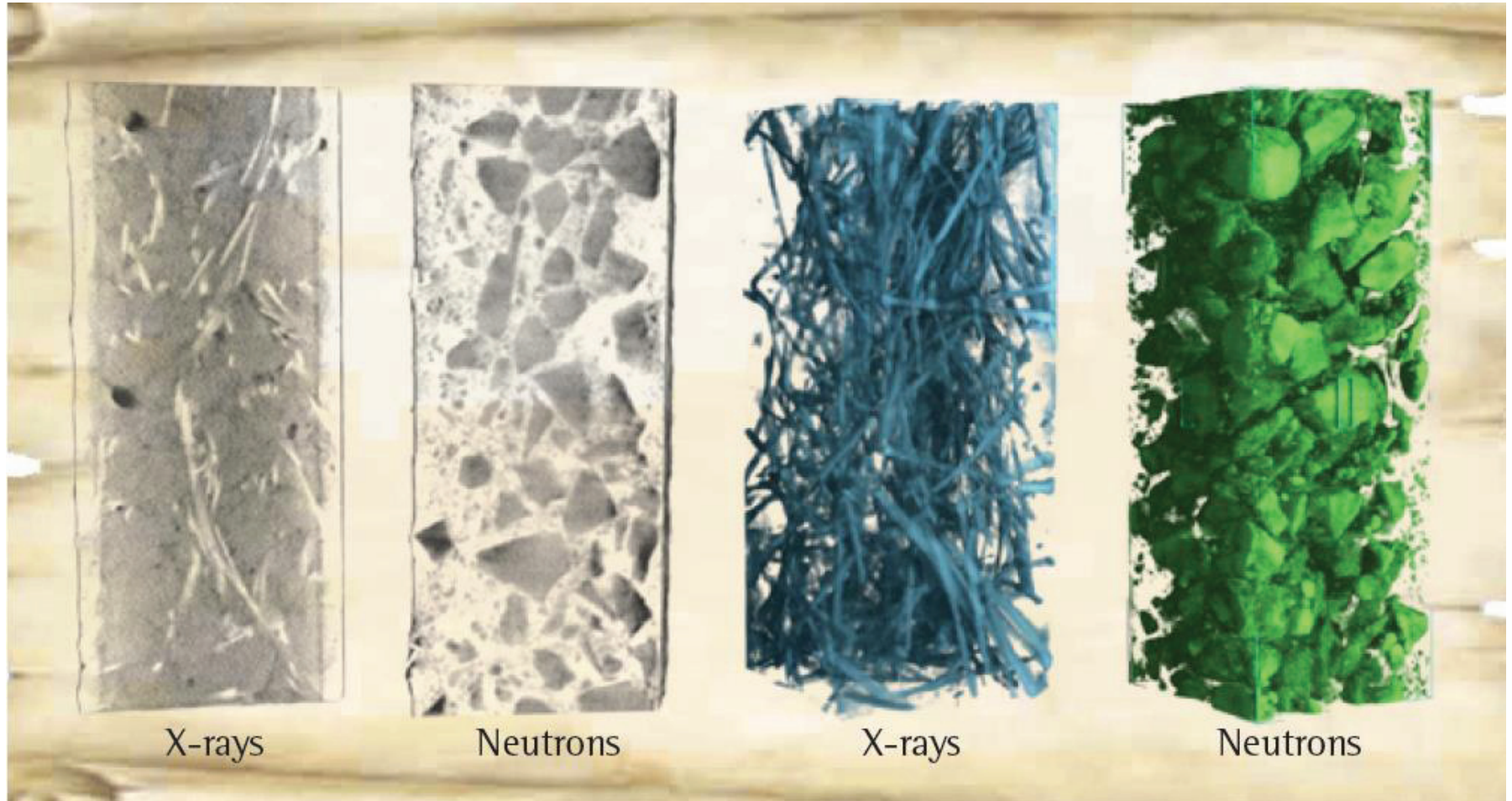


USA 2006:
SNS (<1.4 MW)

Light and neutrons

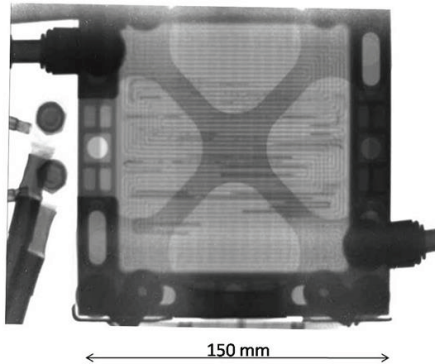


Engineering materials



Properties of neutrons for measurements

Charge neutral
deeply penetrating

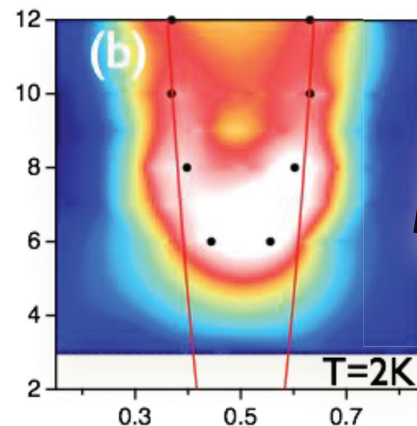


Li motion in fuel cells



Help build electric cars

$S=1/2$ spin
probe directly magnetism



**Solve the puzzle of High-Tc
superconductivity**

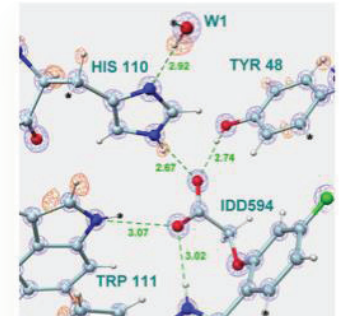


Efficient high speed trains

Nuclear scattering
**sensitive to light elements
and isotopes**

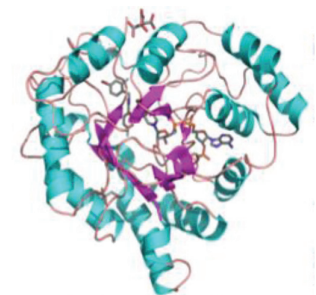


**Test AdS/CFT
correspondence**



Active-site region of hAR

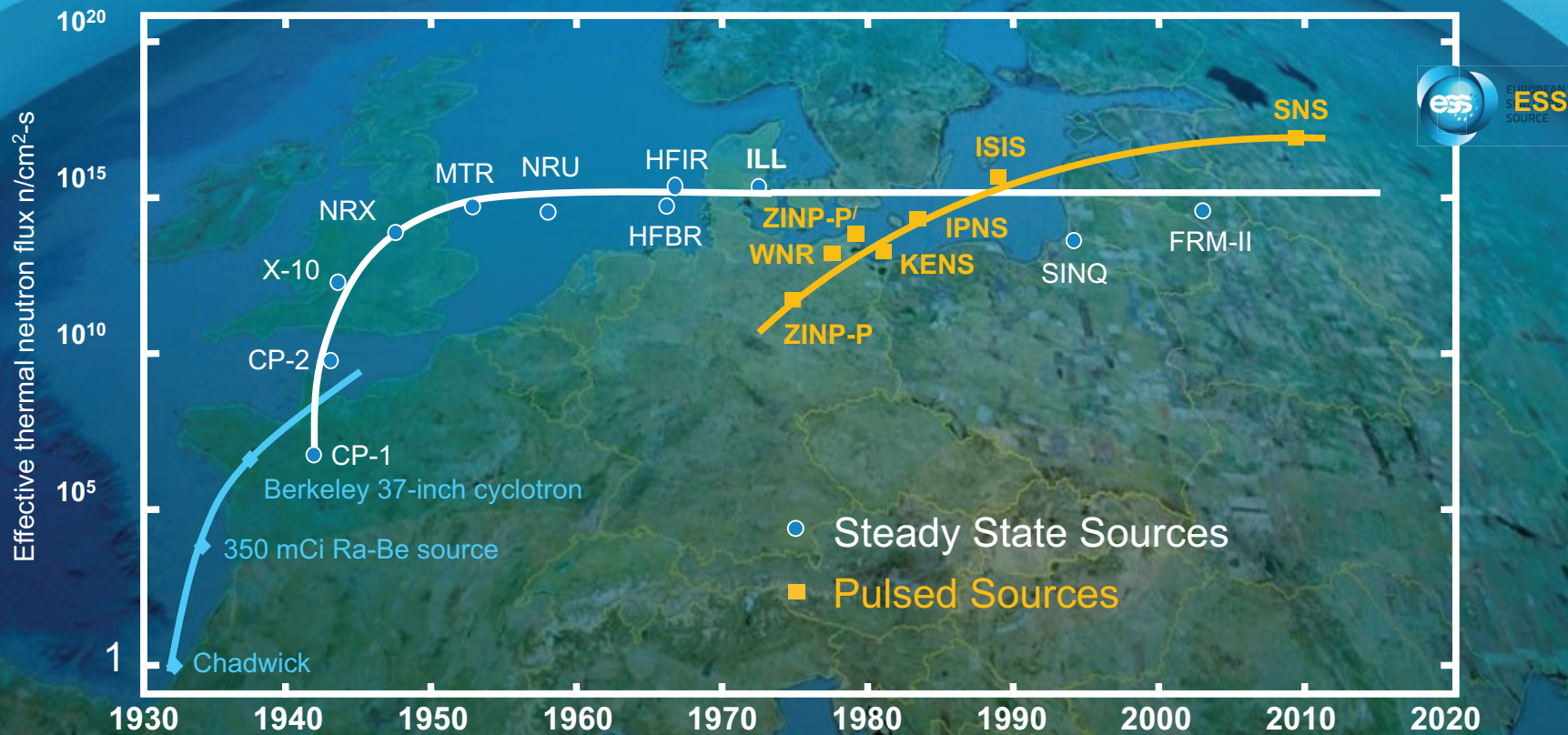
Actives sites in proteins



**Tertiary structure of
hAR**

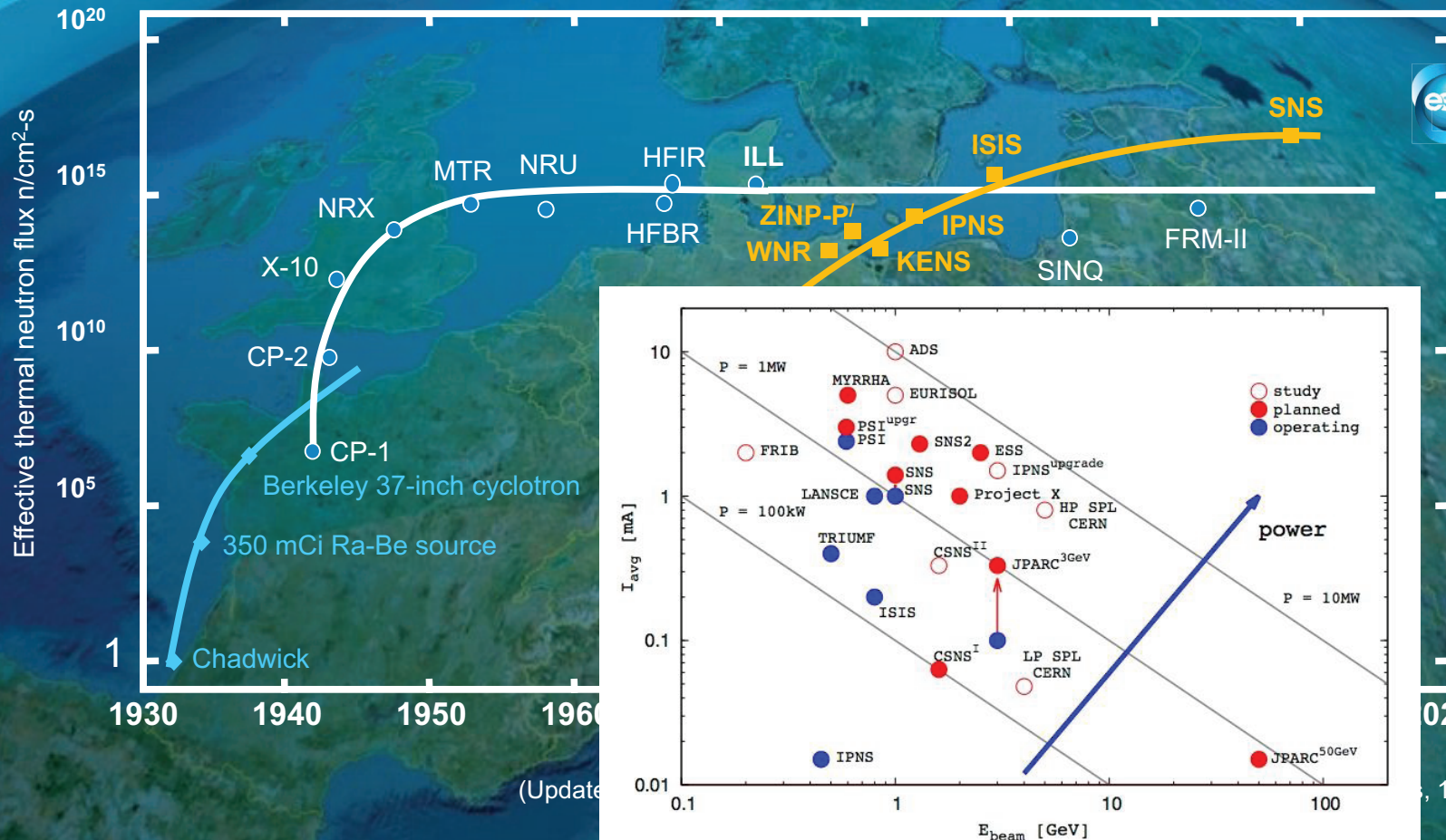
Better drugs

ESS - Bridging the neutron gap



(Updated from *Neutron Scattering*, K. Skold and D. L. Price, eds., Academic Press, 1986)

ESS - Bridging the neutron gap



17 nations committed to build ESS

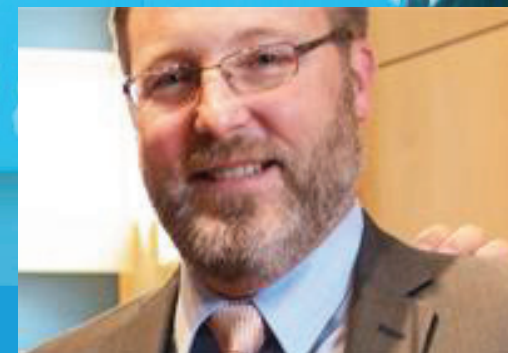


**Cash contributions
from Sweden, Denmark
and Norway**

50% of construction and
20% of operations costs

**In-kind contributions
from the other
14 nations**

Construction cost: 1843 M€
Operation cost: 140 M€
Decommissioning cost: 177 M€



Prototyping the ESS accelerator



Sebastien Bousson



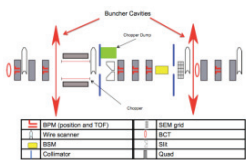
Pierre Bosland



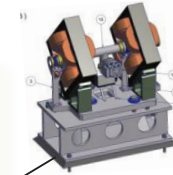
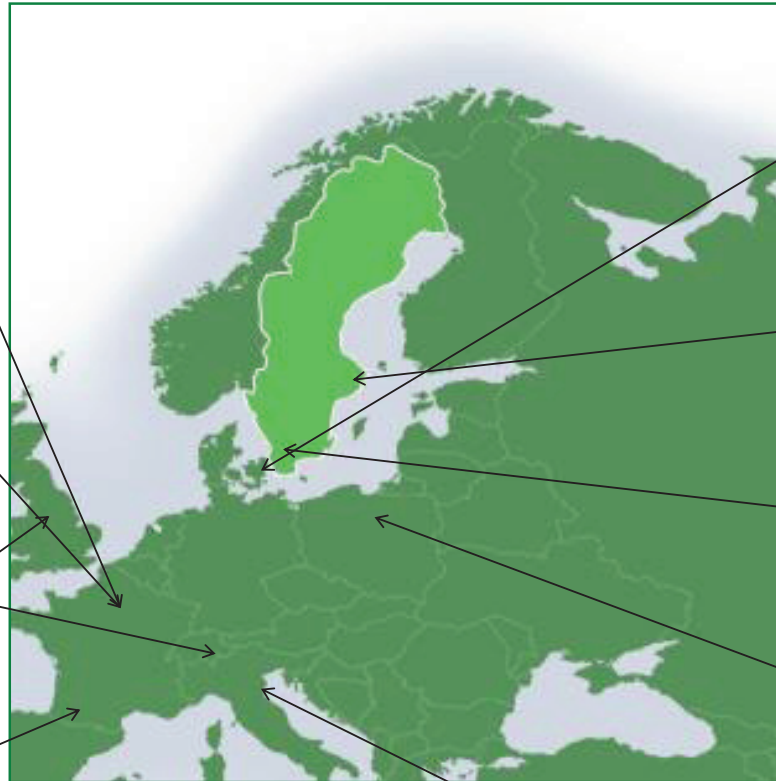
CERN



Roger Barlow



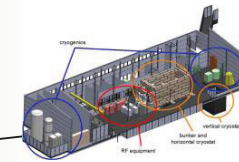
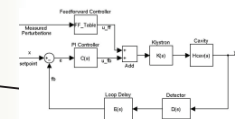
ESS
Bilbao



Søren Pape
Møller

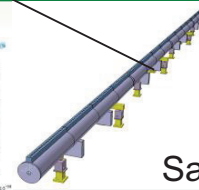
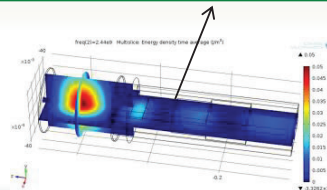


Roger
Ruber

UPPSALA
UNIVERSITET LUNDS UNIVERSITET

Anders J
Johansson

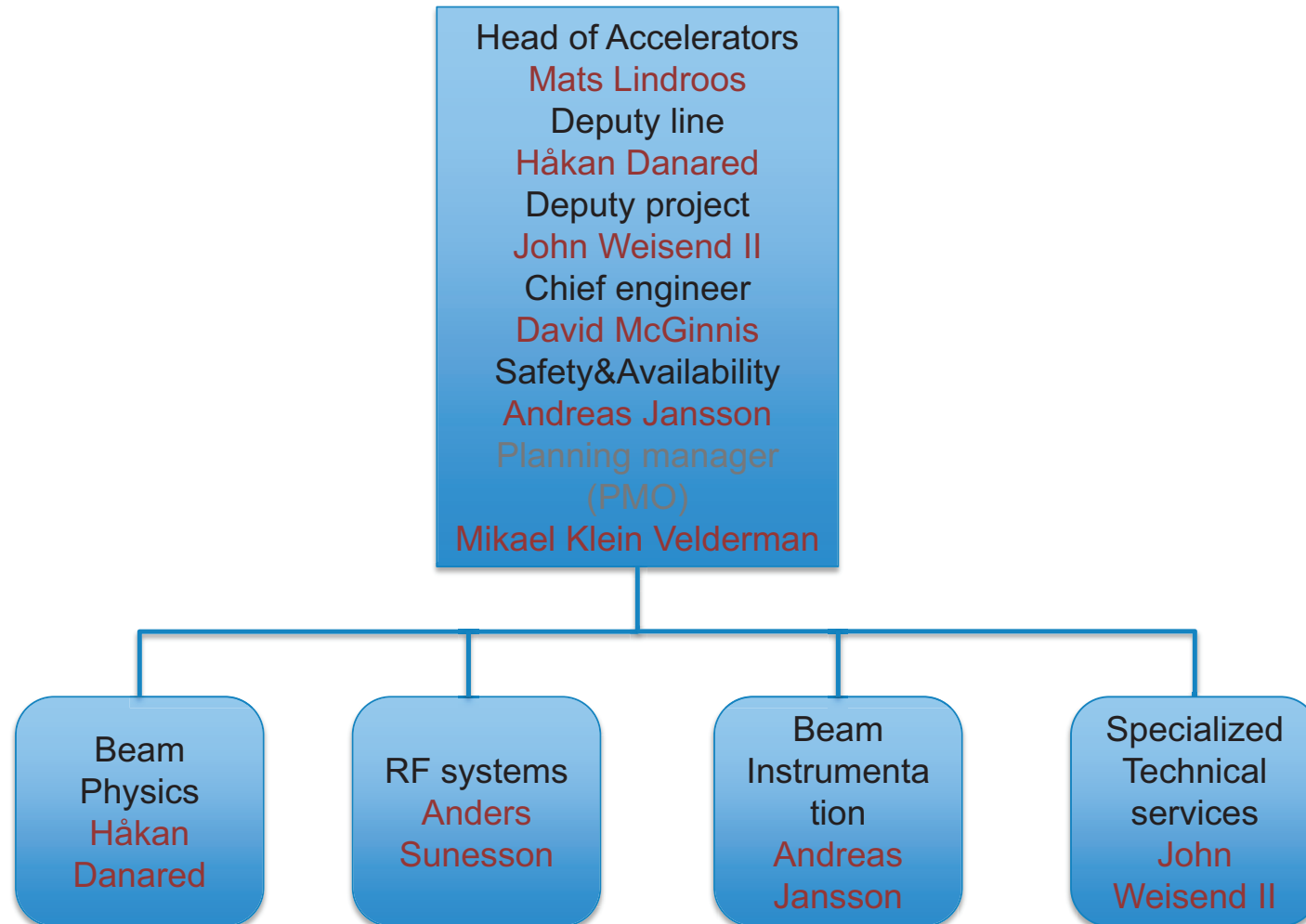
The National Center for
Nuclear Research, Swierk



Santo Gammino



Accelerator organisation



Accelerator organisation

WP#	WP TITLE	WP LEADER	EXTER NAL WP?	LIASON FOR EXTERNAL WP
1	MANAGEMENT	J.G. WEISEND II	NO	
2	ACCELERATOR PHYSICS	M. ESHRAQI	NO	
3	NORMAL CONDUCTING FRONT END	S. GAMMINO	YES	A. PONTON
4	SPOKE CRYOMODULES	S. BOUSSON	YES	S. MOLLOY
5	ELLIPTICAL CRYOMODULES	P. BOSLAND	YES	C. DARVE
6	HEBT & MAGNETS	S. MØLLER	YES	P. LADD
7	BEAM DIAGNOSTICS	A. JANSSON	NO	
8	RF SYSTEMS	A. SUNESEN	NO	
9	ACCEL INFRASTRUCTURE & INSTALLATION	G. LANFRANCO	NO	
10	TEST STANDS	W. HEES	MIXED	W. HEES
11	CRYOGENICS	P. ARNOLD	NO	
12	VACUUM	P. LADD	NO	
13	SAFETY & RELIABILITY	A. JANSSON	NO	
14	REDESIGN EFFORT	D. McGINNIS	NO	
15	COOLING & ELECTRICAL SUPPORT	F. JENSEN	NO	

ESS accelerator

Design Drivers:

High Average Beam Power

5 MW

High Peak Beam Power

125 MW

High Availability

> 95%

Key parameters:

-2.86 ms pulses

-2 GeV

-62.5 mA

-14 Hz

-Protons (H⁺)

-Low losses

-Low heat loss cryostats
for minimum energy
consumption

-Flexible design for
future upgrades



Linac Design Choices

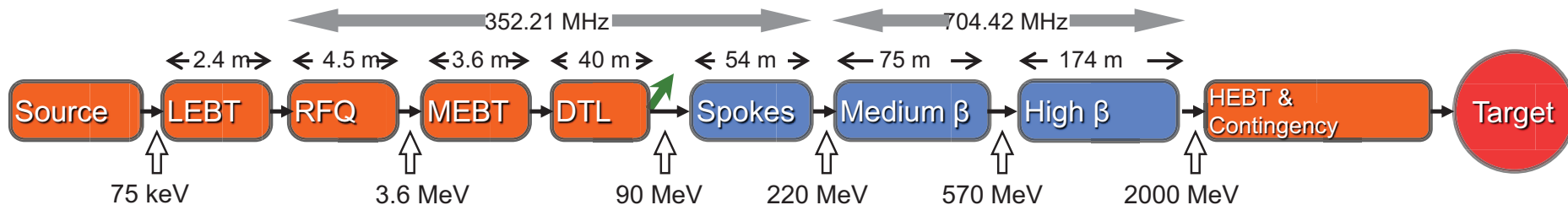
- The energy of the linac is a tradeoff of
 - Linac length
 - Beam current:
 - Space charge forces
 - Halo losses
- Copper Linac
 - Low construction costs but high operational costs
 - Small bore radius < 3 cm
 - Long linac > 750 meters for 2 GeV
- Superconducting Linac
 - High construction costs but low operational costs
 - Large bore radius > 7 cm
 - Short Linac < 360 meters for 2 GeV



2013 Re-design of the linac

- **Redesign undertaken after cost objectives were set autumn 2012 (first costing was a bottom up costing)**
- By removing cryomodules and compensate by using the operational experience from SNS and run the linac aggressively at maximum gradient except in a few extra cryomodules for matching the spokes to the medium beta.
 - **Associated risk:** Decrease of availability and larger beam loss
- By removing high beta cryomodule and compensate by increasing the peak field in the cavities from 40 MV/meter to 45 MV/meter.
 - **Associated risk:** Decrease of availability, cost increase due to drop in cavity yield
- By removing cryomodules by dropping the maximum energy in the linac from 2.5 GeV to 2.0 GeV and compensate by increasing the peak beam current from 50 to 62.5 mA but limit peak power in the RF coupler to 1100 MW.
 - **Associated risk:** Decrease of availability and larger beam loss
- Implement energy staging. Complete remaining installation of linac to 2.0 GeV to provide 5 MW of beam by 2022.
- By introducing new MB IOT technology for the high beta cryomodules in the baseline.
 - **Associated risk:** R&D not finished in time

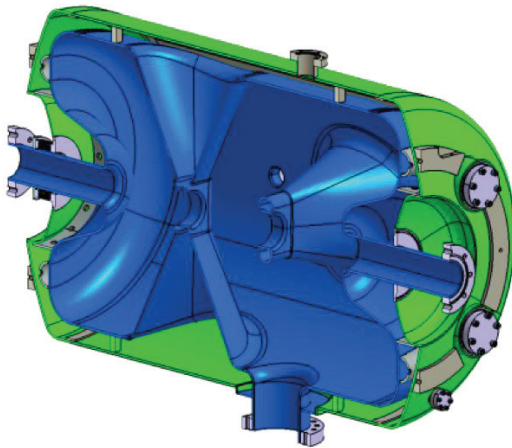
ESS Linac



	Energy (MeV)	No. of Modules	No. of Cavities	βg	Temp (K)	Cryo Length (m)
Source	0.075	1	0	—	~300	—
LEBT	0.075	—	0	—	~300	—
RFQ	3.6	1	1	—	~300	—
MEBT	3.6	—	3	—	~300	—
DTL	90	5	5	—	~300	—
Spoke	220	13	2 (2S) $\times$ 13	$0.5 \beta_{opt}$	~2	4.14
Low β	570	9	4 (6C) $\times$ 9	0.67	~2	8.28
High β	2000	21	4 (5C) $\times$ 21	0.86	~2	8.28
HEBT	2000	—	0	—	~300	—

Cavities

- Spoke cavities at IPNO
 - A double spoke cavity, the coupler, helium vessel, ancillaries and cryomodule.
 - Cavity made of 4 mm thick bulk niobium
 - 4 HPR ports on one end
 - Helium vessel made of 4 mm thick titanium
 - Coupler is the EURISOL design
 - The production of 3 prototypes had started in March 2013



- Elliptical cavities at CEA
 - 2 high- β prototyped
 - Medium β design completed

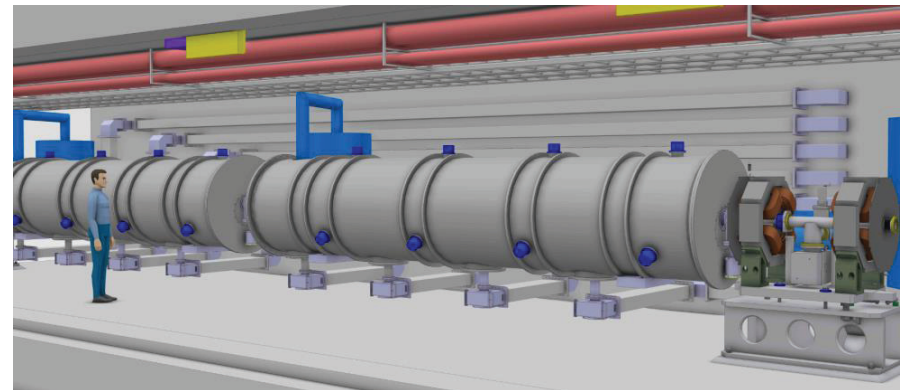
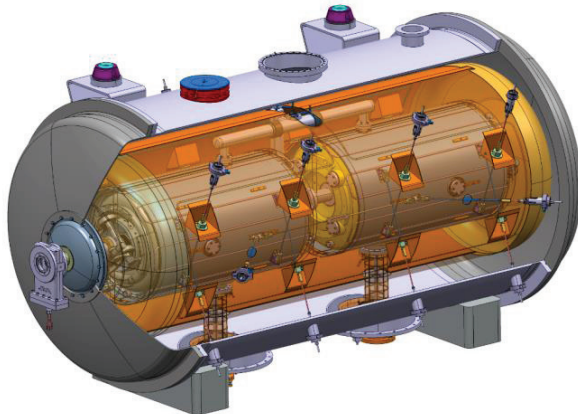
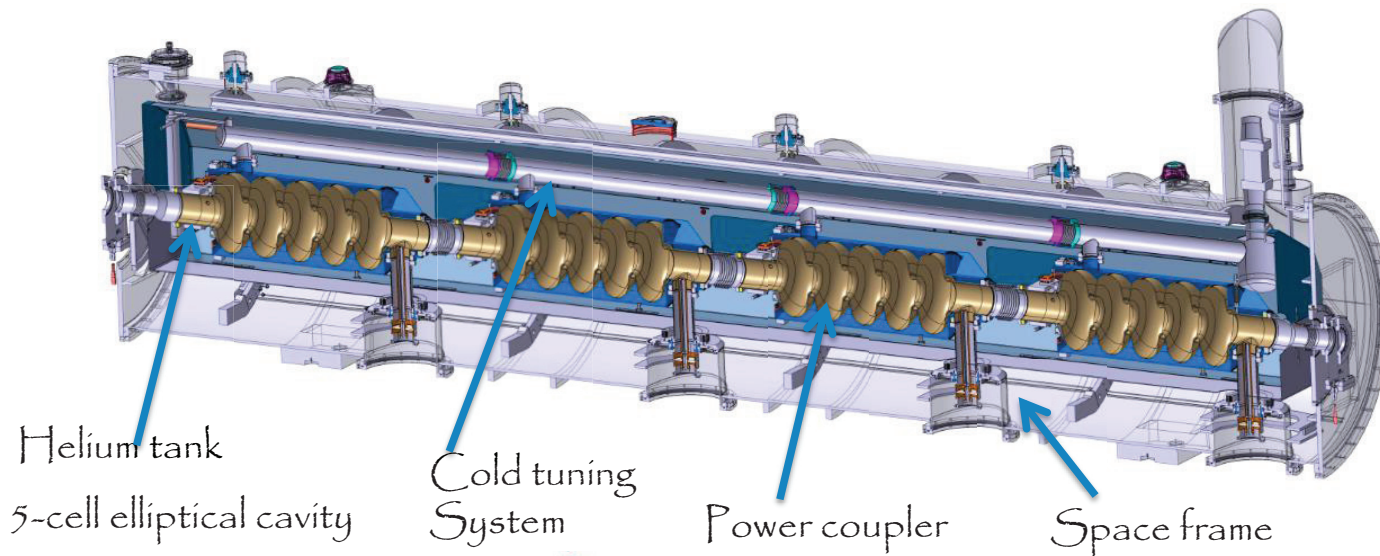


First born! $\beta=0.86$

β_g	E_{acc} (MV/m) Ver. Test	E_{acc}^* (MV/m) linac	$Q_0 @$ E_{acc}	N Cells
0.67	17	16.7	5×10^9	6
0.86	20	19.9	6×10^9	5

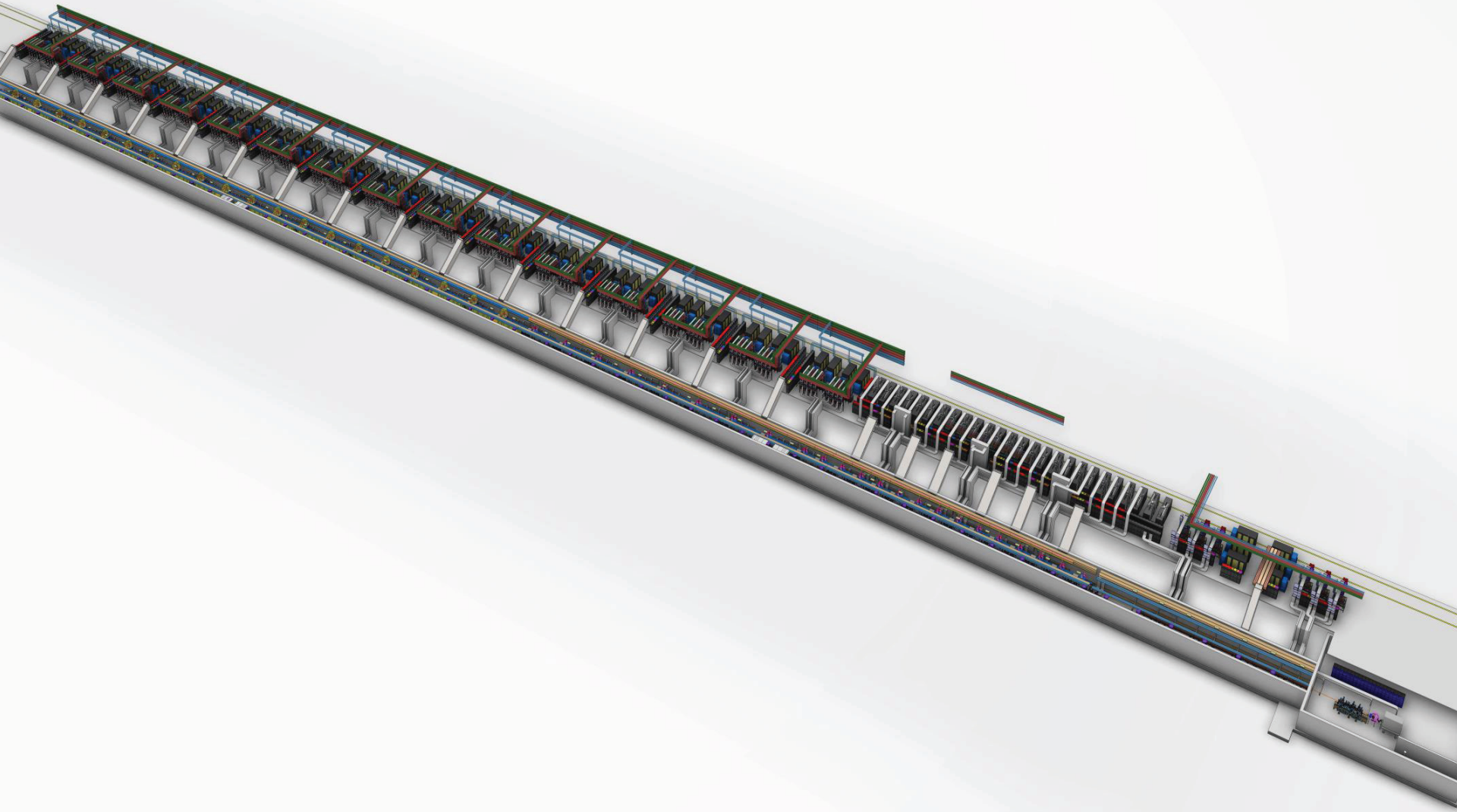
* New design with higher peak surface field!

Cryomodules Technology demonstrators

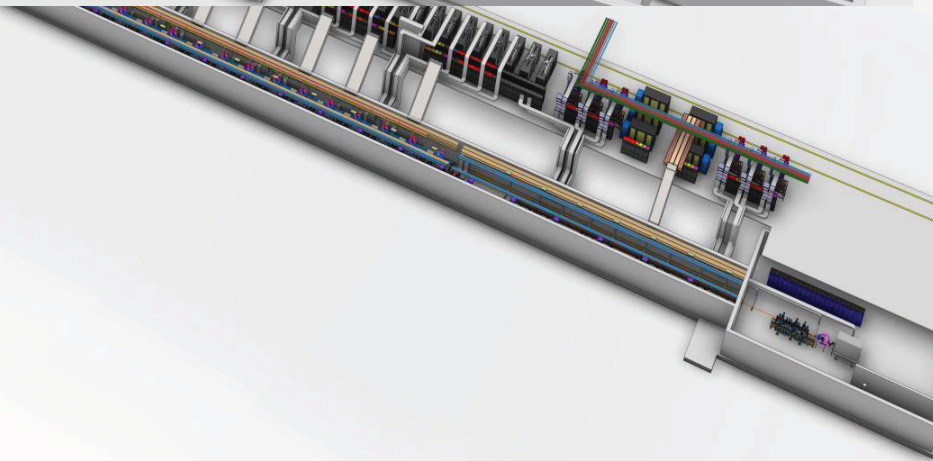
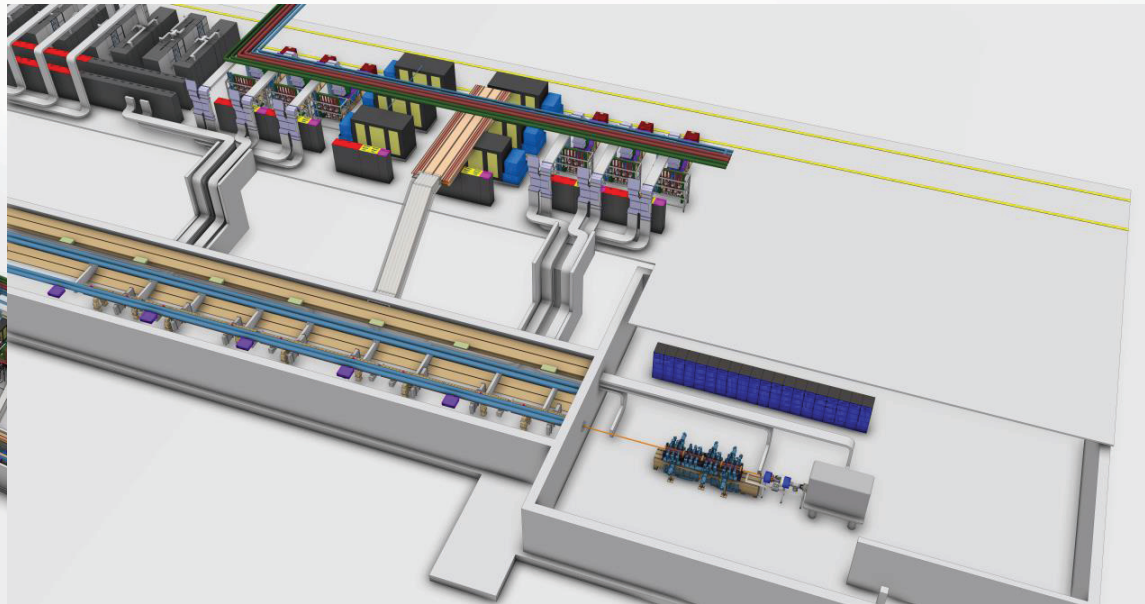
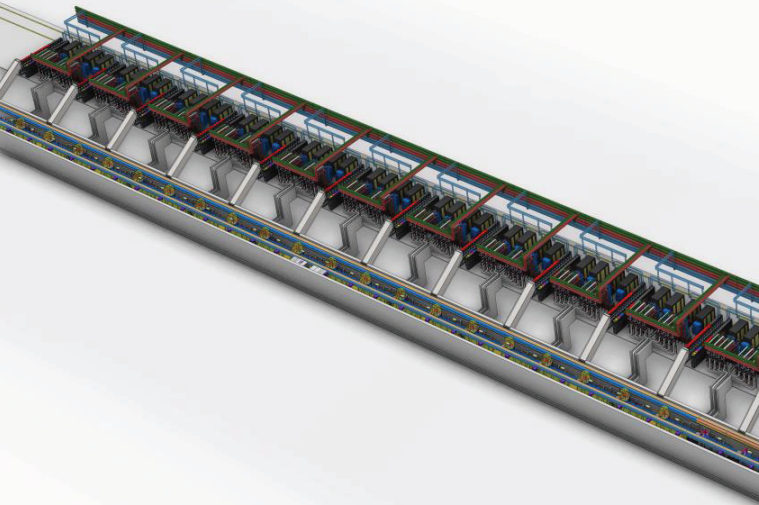


→ Cavities Cryomodule Technology Demonstrator results by the end of 2015

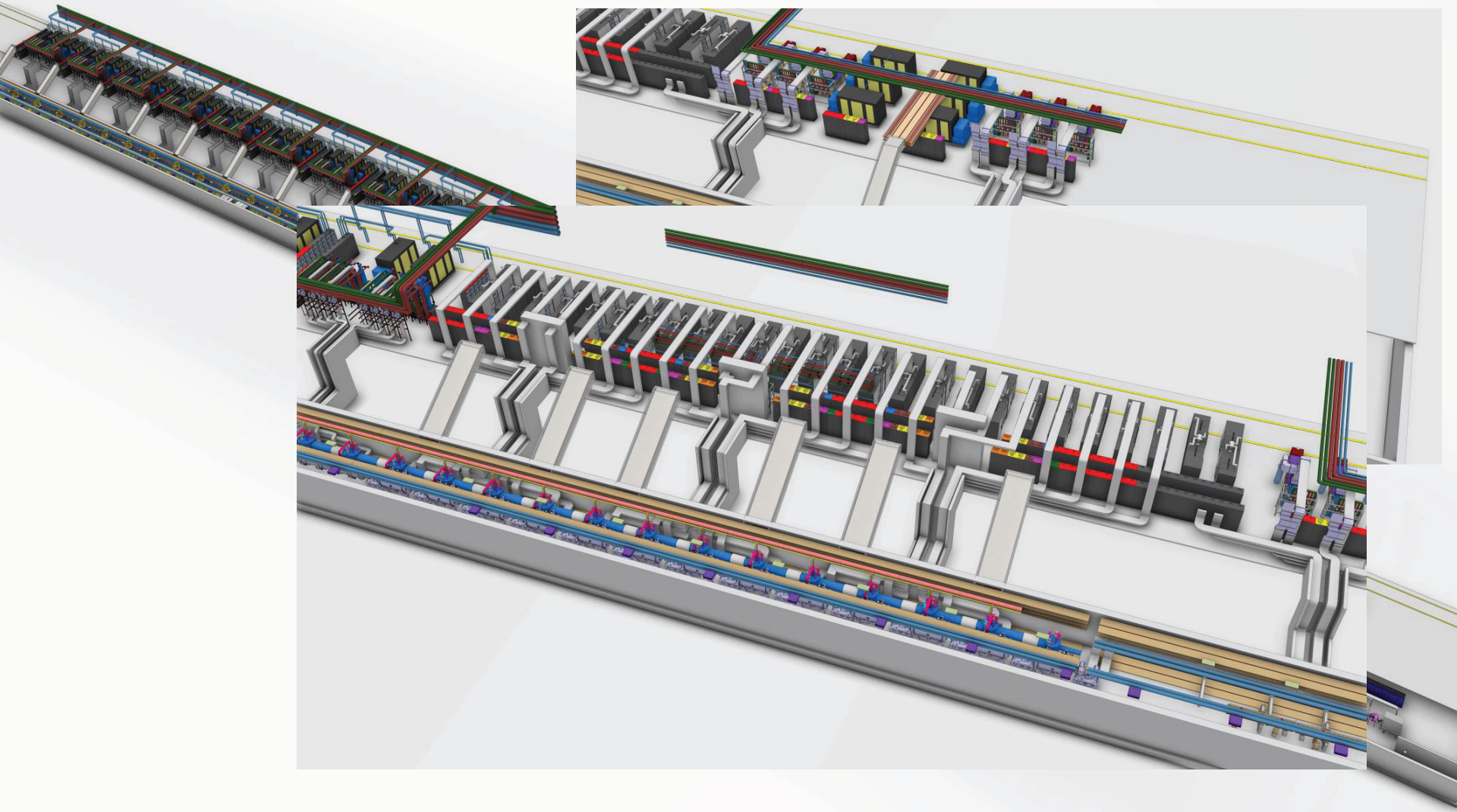
Linac layout



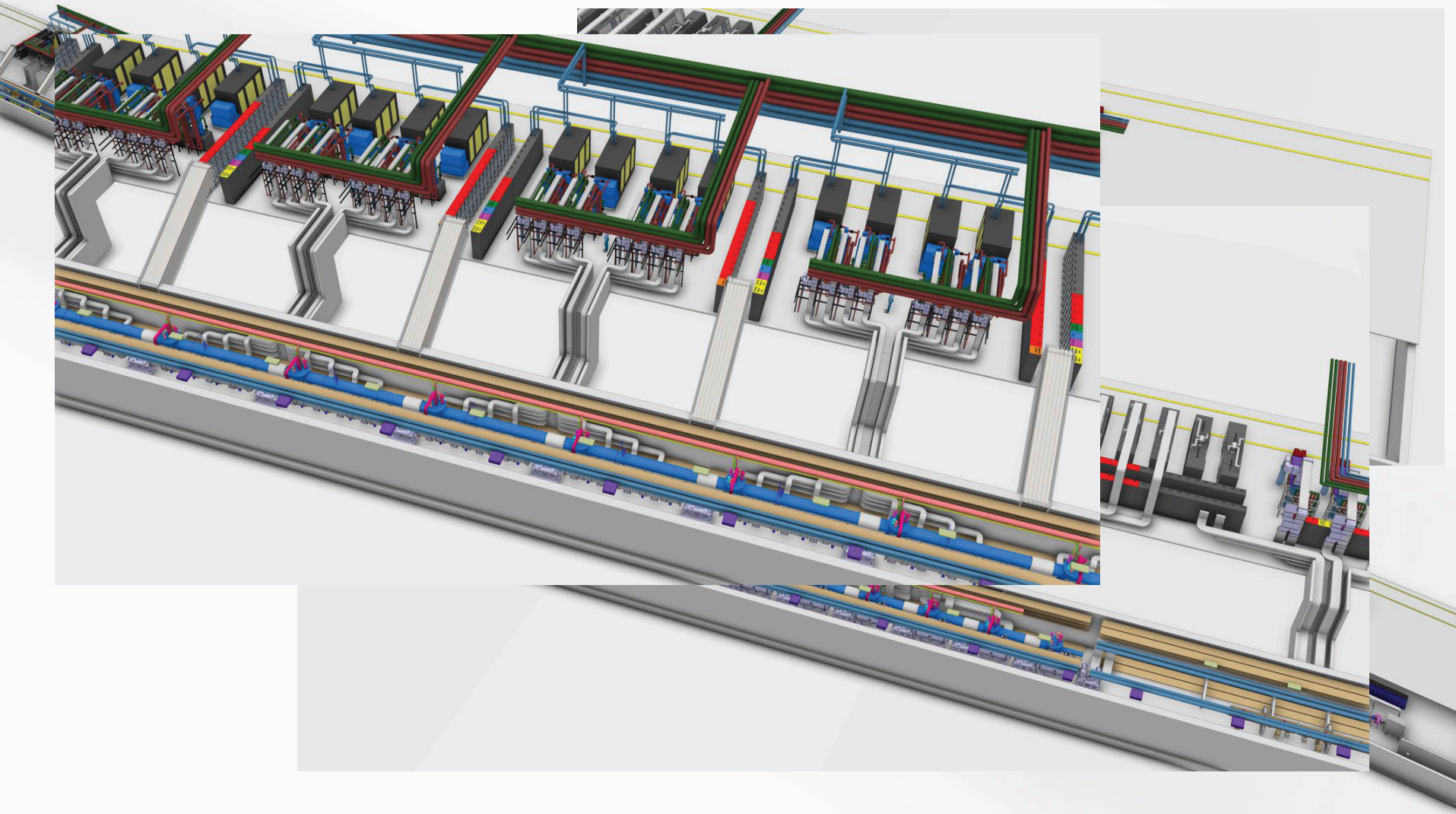
Linac layout



Linac layout

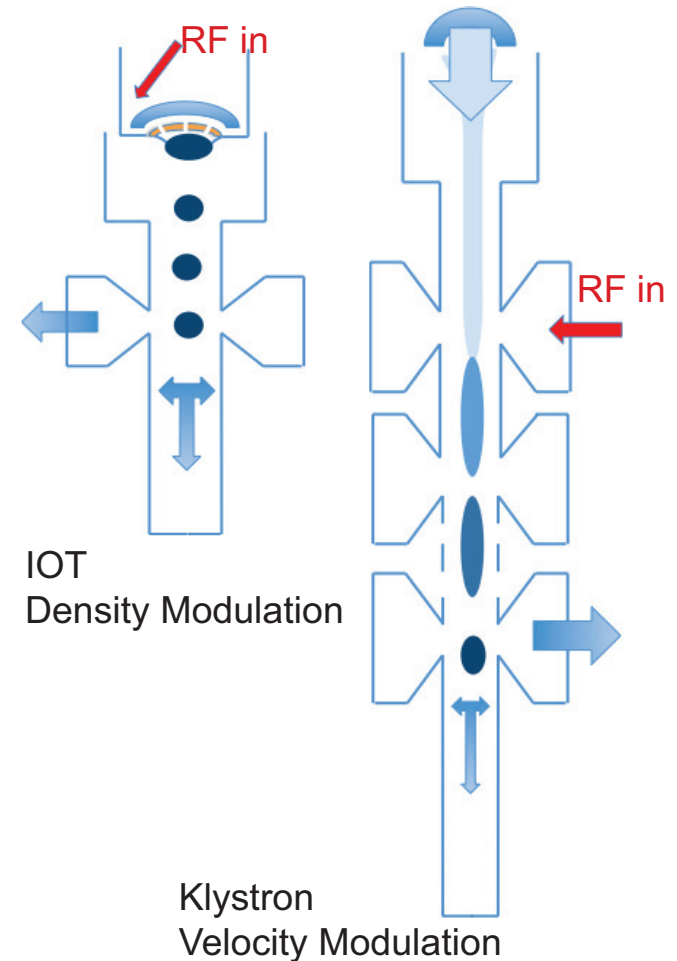


Linac layout



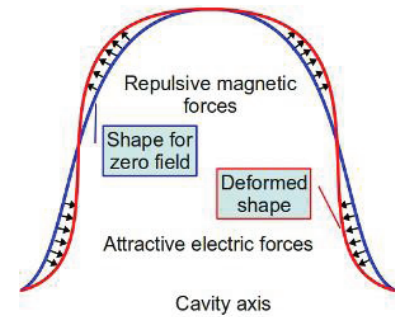
Inductive Output Tubes or klystrons?

- ESS
 - Induction Output Tubes, IOTs
 - Higher electrical efficiency
 - They don't conduct in the absence of input drive
 - Compact
 - Short MTTR
 - Cheaper modulator (No high voltage switching)
 - Why suddenly IOTs?
 - Development of Pyrolytic graphite grids
 - Solid state drivers

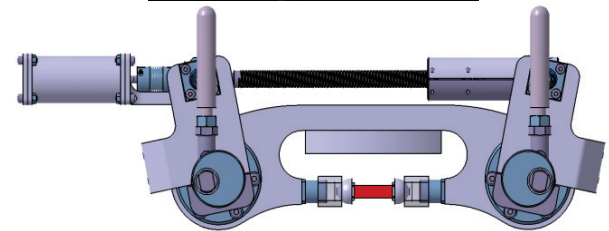


Lorentz De-tuning

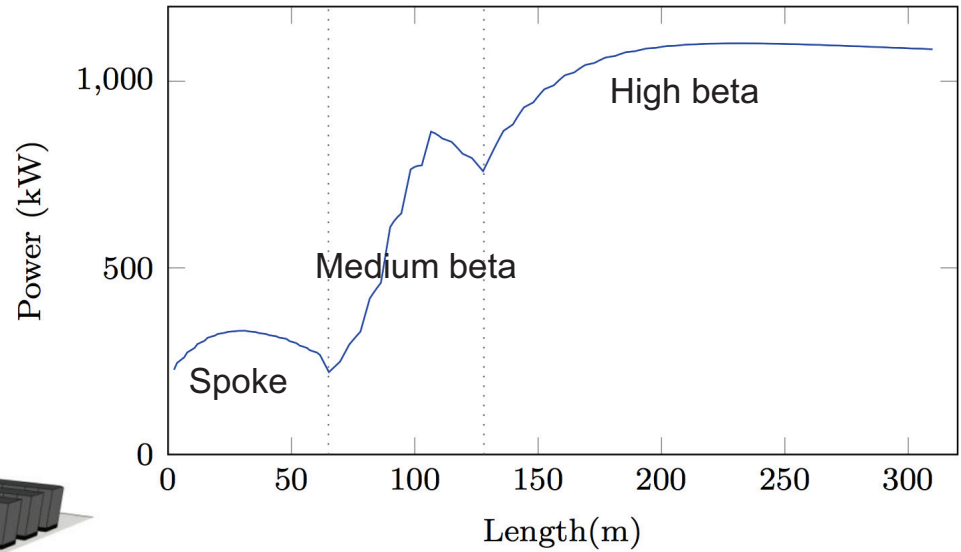
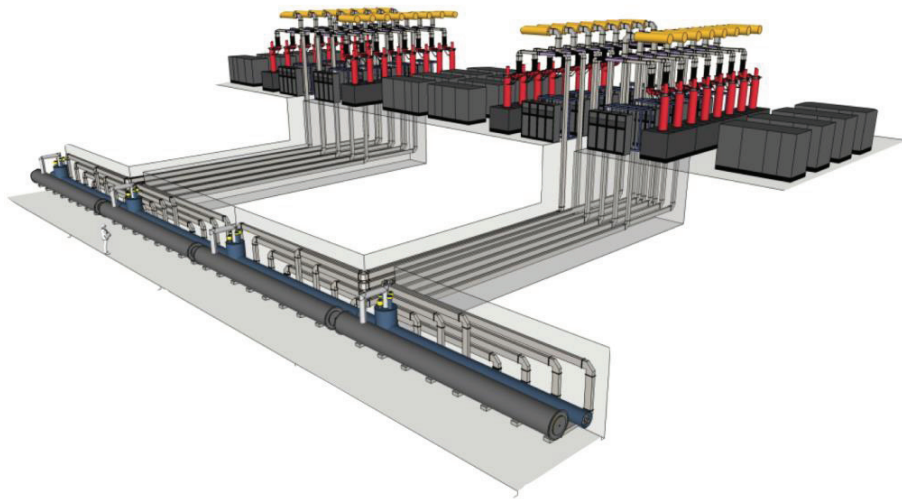
- We expect over 400 Hz of detuning in the ESS cavities.
 - Unloaded cavity bandwidth = 0.07 Hz
 - Loaded cavity bandwidth = 1 kHz
- The mechanical time constant of the cavities is about 1 ms compared to the pulse length of 3 ms
 - Static pre-detuning as done in SNS will not be sufficient
 - Dynamic de-tuning compensation using piezo-electric tuners is a must!
 - Or else pay for the extra RF power required

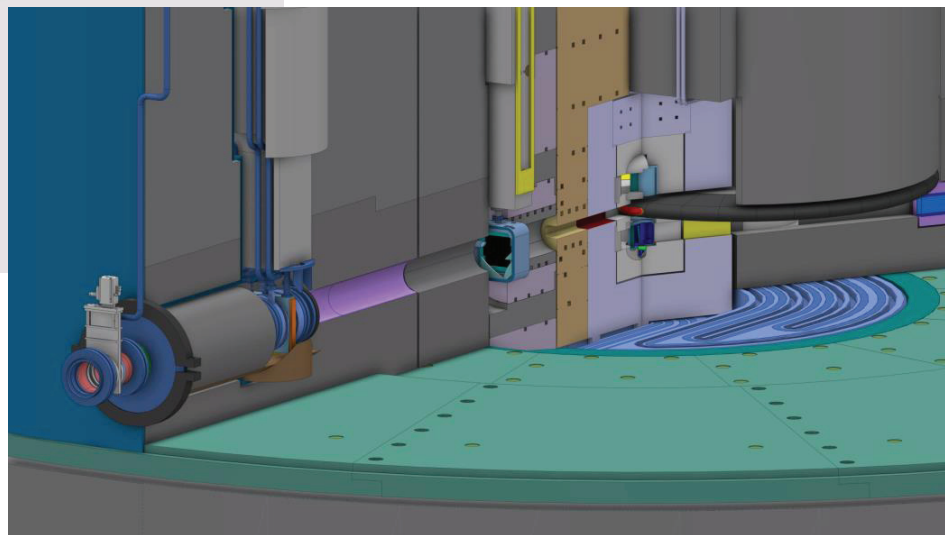
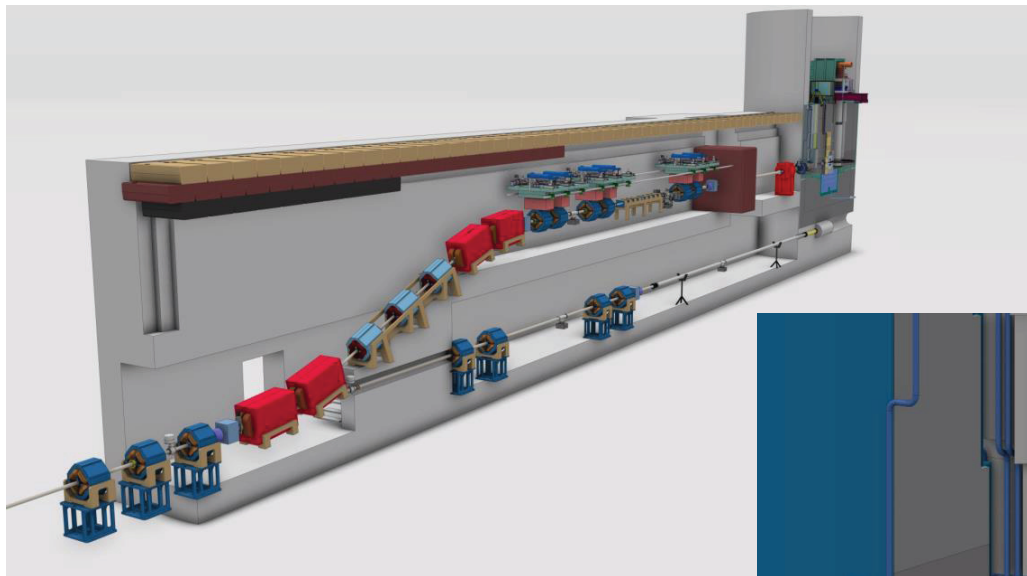


$$P_s = \frac{1}{4} (\mu |\vec{H}|^2 - \epsilon_0 |\vec{E}|^2)$$
$$\Delta f_0 = (f_0)_2 - (f_0)_1 = -K E_{acc}^2$$



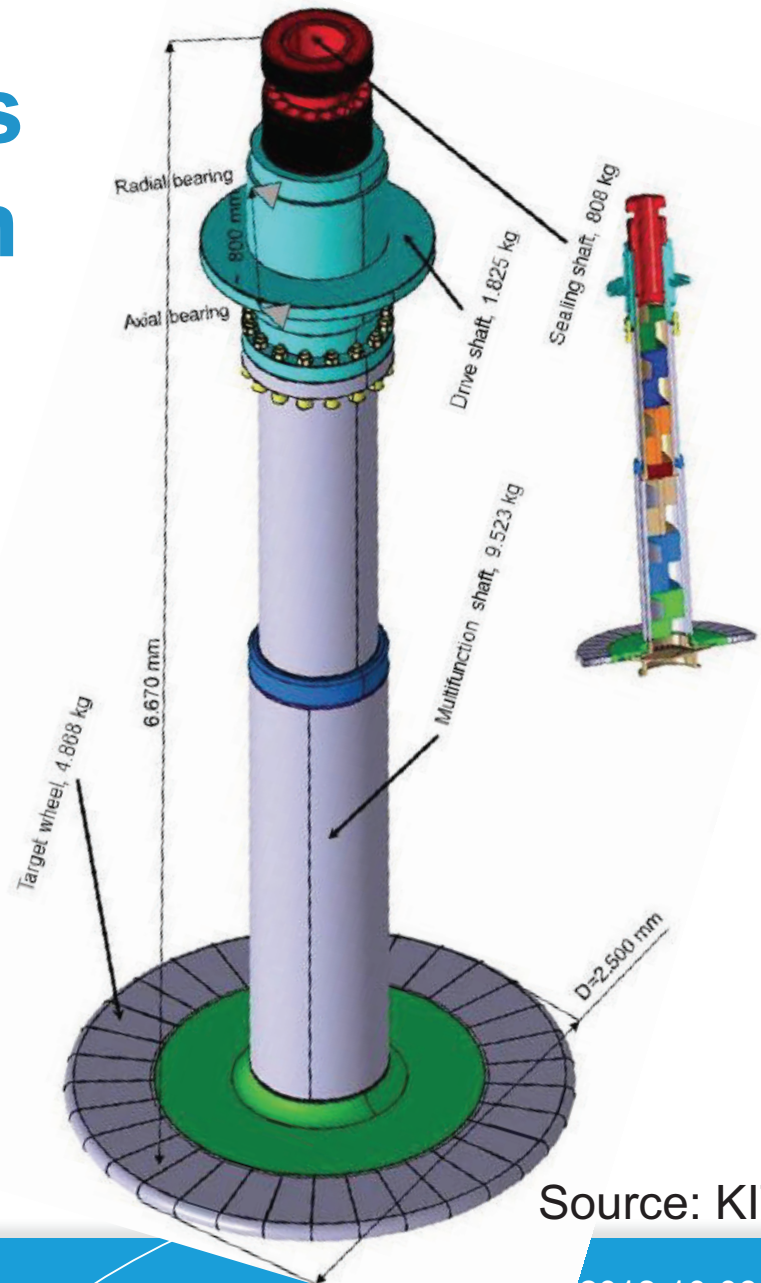
RF is all!





High beam power leads to an innovative design

- Rotating tungsten target
 - 2.5 m diameter
 - 33 sectors
 - Suspended on a 6 m shaft
 - Total mass: 17 tonnes
 - Rotational speed = 0.42 Hz
 - Estimated lifetime = 5 years
- Helium coolant
 - 3 kg/s flow rate
 - 4 bar inlet, 1 bar ΔP
 - 20°C inlet, 200°C ΔT



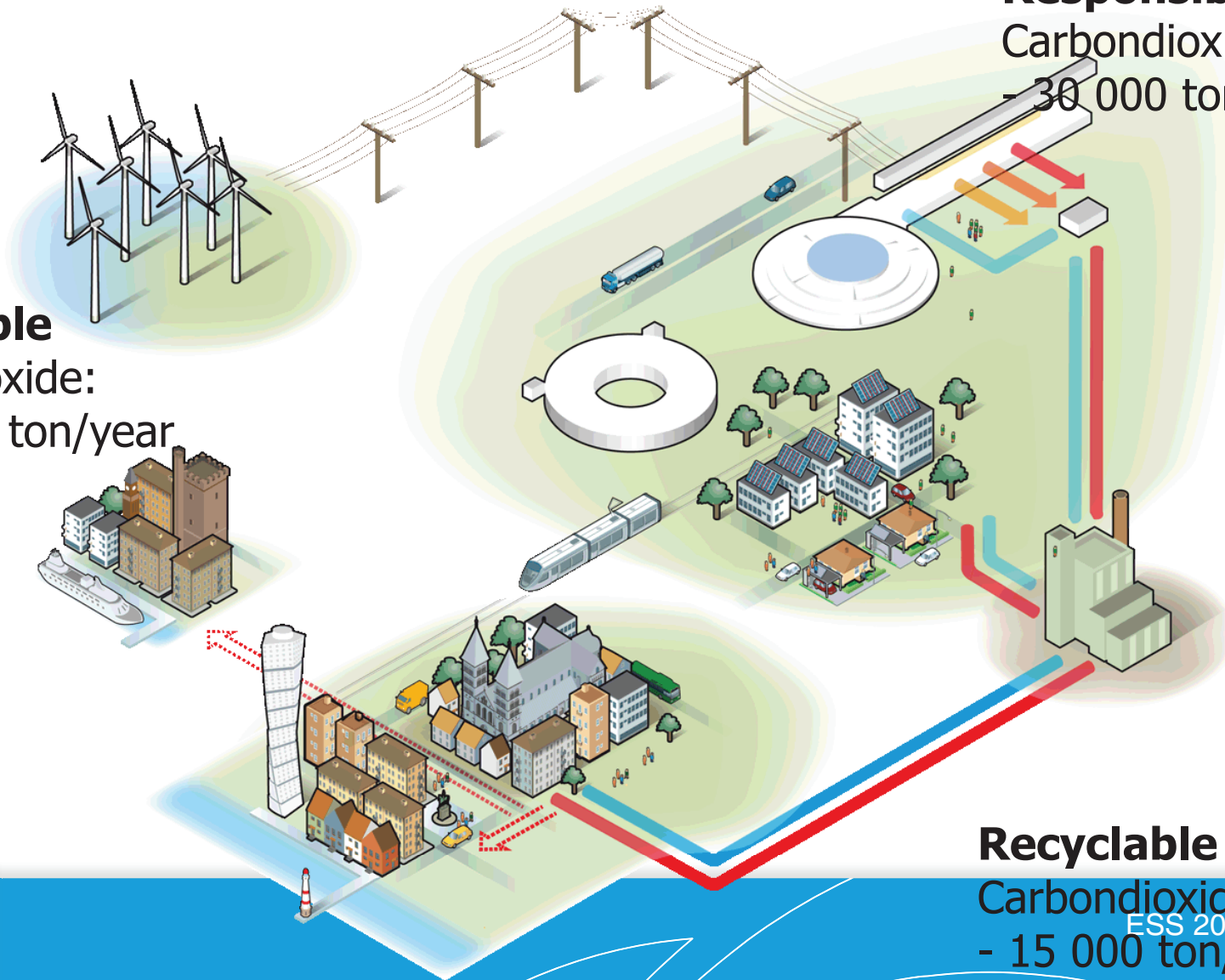
Source: KIT

A sustainable research facility

Renewable
Carbondioxide:
- 120 000 ton/year

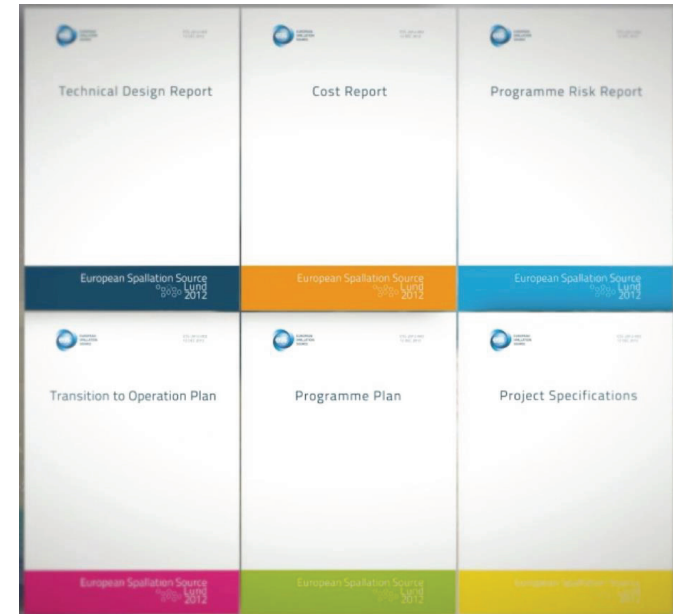
Responsible
Carbondioxide:
- 30 000 ton/year

Recyclable
Carbondioxide:
- 15 000 ton/year



In-kind Process for Construction Begins

- Call for **Expressions of Interest**
- Based on TDR, Project Plan & Cost Book
- May contribute “component” or “work”
- Several calls; WPs will be released as they mature
- Slightly different for Accelerator, Target & Instruments
- Competence of the team(s) responding, references
- We expect there will be partnering on Work Packages
- An Institution or Laboratory, Company, or a Combination
- After EoI response, a detailed discussions for IKC begin



- Allen Weeks
Head of Communications @ ESS

Headlines

- The ESS Accelerator Design answers ESS needs, it holds a lot of potential for a long life with many upgrades – maybe even a neutrino beam...
- The ESS partners (collaborations) are essential for the project
- Experience from SNS has been very important for design, costing and planning
- A large part of the contributions will come as in-kind from the member states
- We are building ESS together!



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Welcome

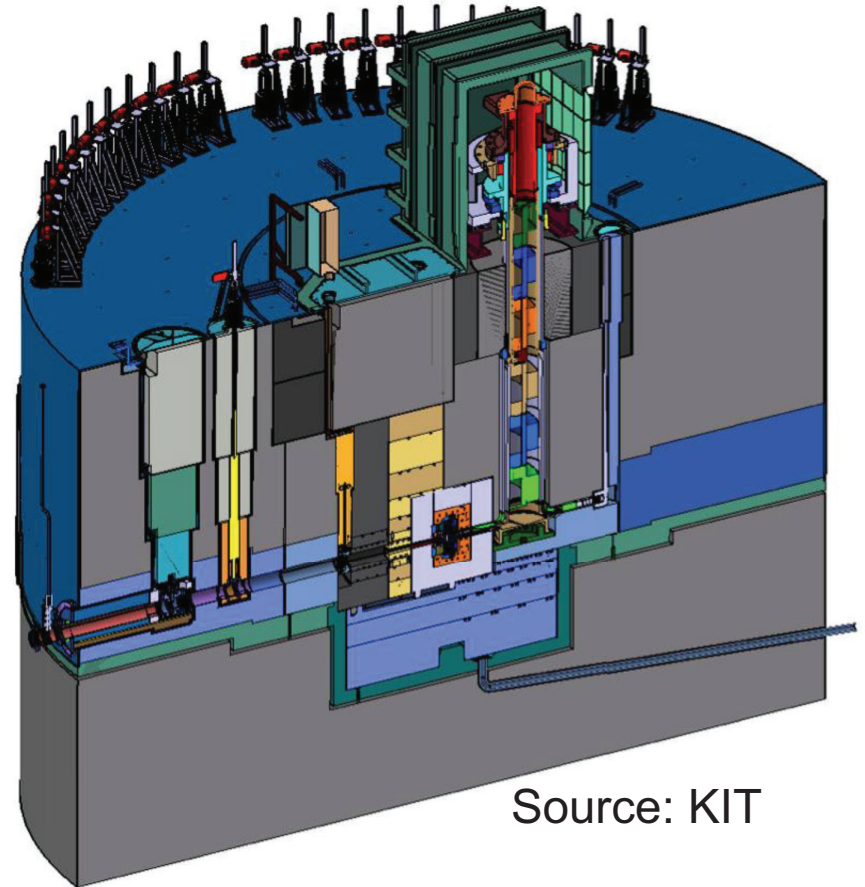
ESS 2013-10-06

Collaborative projects

- ESS is an emerging research laboratory with (still) very limited capacity in-house
- Two possibilities:
 - Limit the scope of the project so that it can be done with in-house resources
 - Work in a collaboration where the scope of the project can be set by the total capacity (distributed) of the partners
- The accelerator part of the project well suited for this as this community has a strong tradition of open collaboration (XFEL, FAIR, CERN-LINAC4, European commission framework programs such as EUCARD and TIARA, EURISOL,...)
- To keep cost down and to optimize schedule this requires that investments in required infrastructure is done at the partner with best capacity to deliver

The target station design is detailed in the Technical Design Report

- Helium-cooled rotating tungsten target
- Nearly 5 MW of heat carried away by cooling systems
- Entombed in 5 m of steel
- Critical components become highly activated, requiring remote handling for servicing

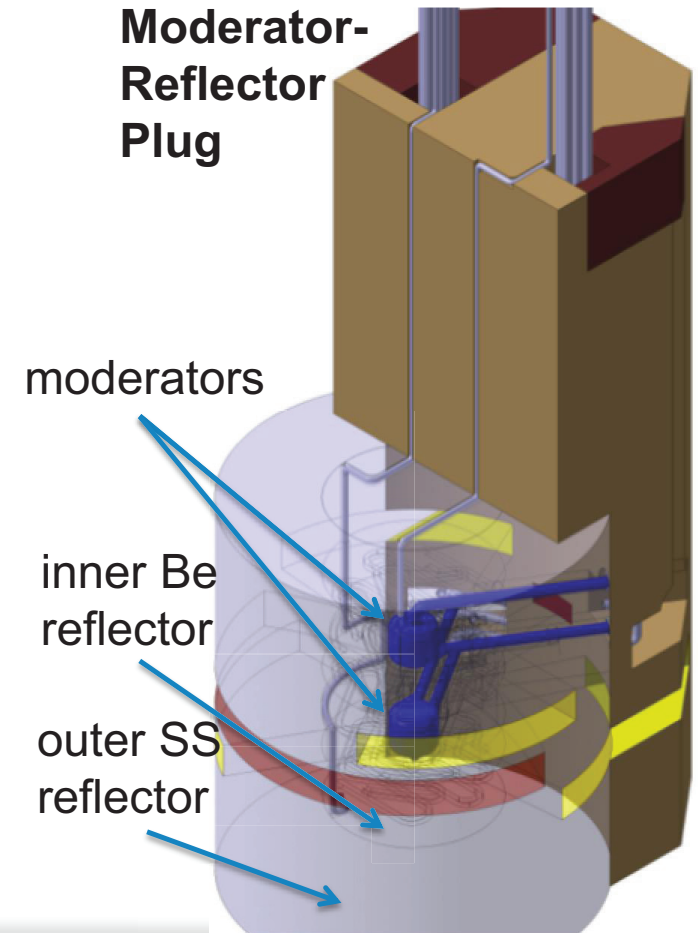


Source: KIT

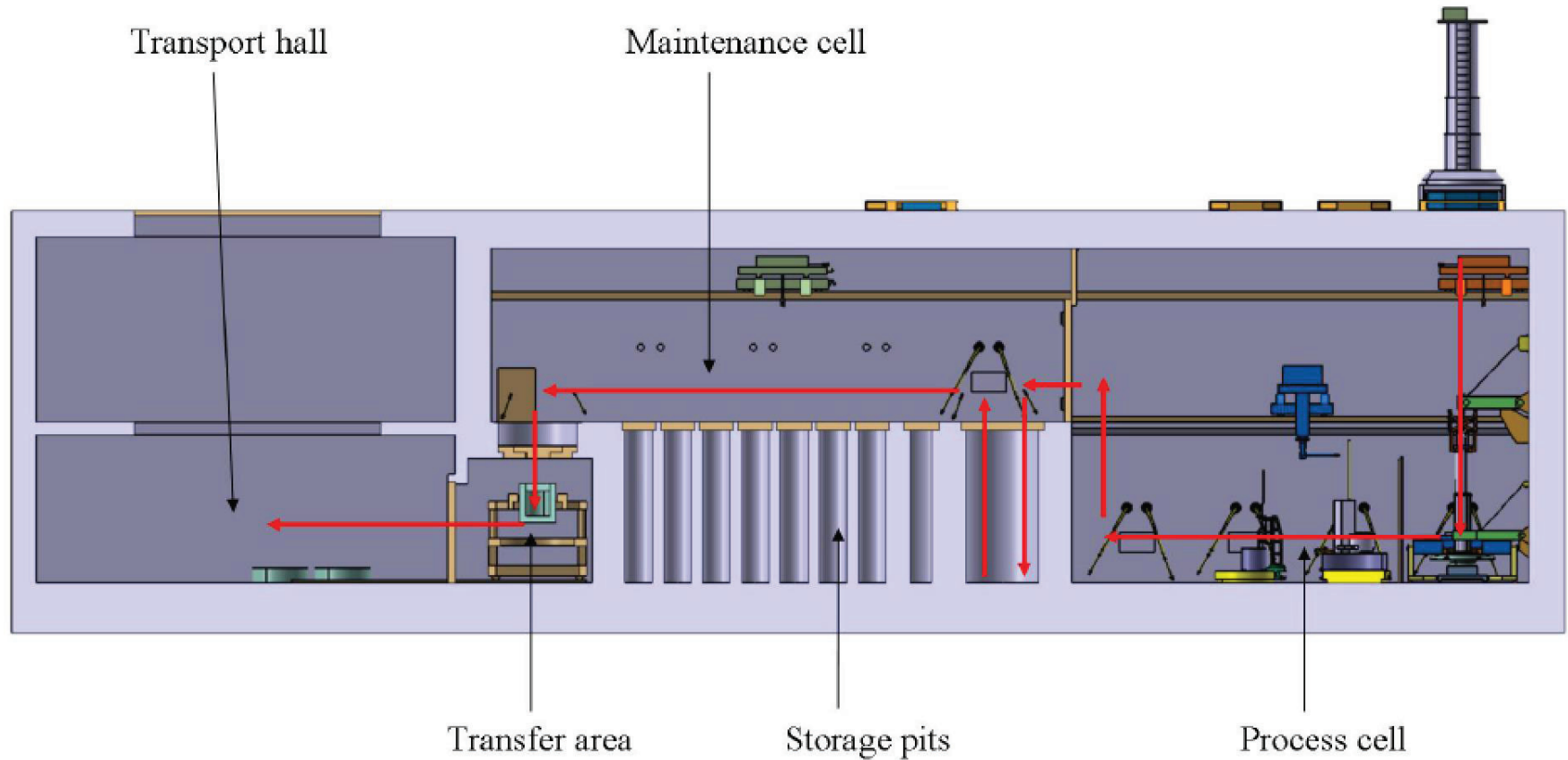
ESS Technical Design Report, ESS-2013-001, April 22, 2013, available at <http://eval.esss.lu.se/cgi-bin/public/DocDB/ShowDocument?docid=274> .

Liquid hydrogen moderators deliver cold neutrons to instruments

- Liquid para-hydrogen at 20 K carries away 20 kW of deposited heat
- Moderators surrounded by beryllium reflector provides peak neutronic performance
- Moderator-reflector plug weighs 24 tonnes
- Estimated lifetime = 1 year

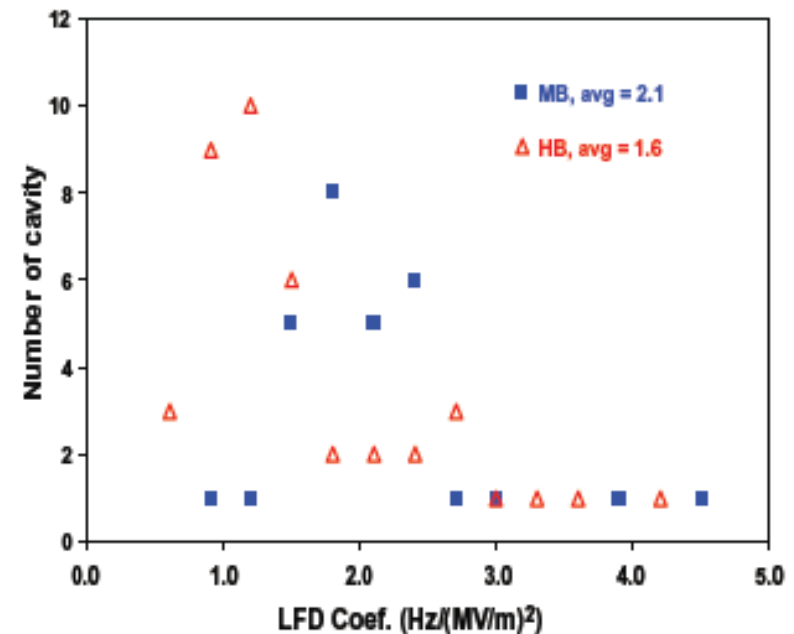


Activated components are handled remotely in hot cells



Cavity Power Configuration

- Because of fabrication techniques,
 - superconducting cavity strings are usually much shorter (< 1 m) than copper cavity strings (> 5 m).
 - The Lorentz de-tuning coefficient varies from cavity to cavity
- Therefore, each superconducting cavity has its own RF power source



Transit Time Factor

- For proton linacs using copper RF cavities
 - the cavity cell structure is tuned to match the changing proton velocity as it accelerates.
 - The power profile is usually flat
- Because of high fabrication costs and difficulty,
 - The cell structure of superconducting cavities is tuned for only one beam velocity.
 - Multiple families of cell velocities are chosen. ESS cell velocities:
 - Spoke: $\beta_g = 0.5$
 - Medium beta: $\beta_g = 0.67$
 - High beta: $\beta_g = 0.86$
 - There is a limit on the surface field in a SCRF cavity (ESS 45 MV/m)
 - Since, the particle velocity does not match the geometrical velocity for the entire acceleration range,
 - **The power profile is not flat**